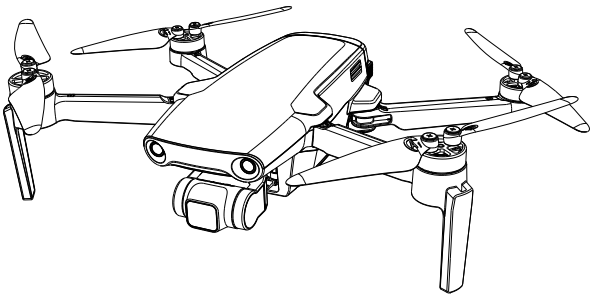


MINI 7

《User Guide》

V26.05.14

INFORMATION TO THE USER

All users must read product operating instructions and liability disclaimer before using and operating any HUBSAN products. By using a HUBSAN product(s), users accept the terms and conditions of HUBSAN liability and operational guidelines. This product is not suitable for minors under 14 years of age. When operating a HUBSAN product(s), users also accept all liability and responsibility for their own behavior, actions as well as any consequences resulting thereof while using a HUBSAN product(s). These products may only be used for purposes that are in accordance with local laws and regulations, and any applicable policies / guidelines HUBSAN may make available. Users agree to comply with these terms and conditions, along with any and all relevant policies / guidelines set forth by HUBSAN. Part of the details of this document may change with the upgrade of the product software version. Please read the upgrade details carefully before upgrading the software version.

INSTRUCTIONS

Some of the product flight functions are restricted in certain areas. Once you use this product, you must strictly read and follow the relevant ICAO regulations, local airspace and UAV regulations. You are liable for any non-compliance with the foregoing, and are responsible for the consequences of your actions as well as any indirect and / or direct liability that arises as a result of these limitations.

FLIGHT ENVIRONMENT REQUIREMENTS

(1) Select an open area to fly. Avoid high rise buildings and tall obstacles (such as trees and poles). Near buildings and obstacles, flight control signals and GPS signals are severely affected; GPS functions such as GPS mode and Return to Home may not function properly.
(2) Do not fly in bad weather conditions (such as in wind, rain or fog).
(3) Fly the aircraft in ambient temperatures of 0-40 ℃.
(4) When flying, please stay away from obstacles, people and crowds, high voltage lines, trees, water, etc.
(5) To avoid remote control signal interference, do not fly in strong electromagnetic environment (such as venues with radio stations, power plants and Phone antenna and TV towers).
(6) The aircraft cannot be used at or near the Arctic circle or Antarctica.
(7) Do not fly in no-fly zones.
(8) Do not operate the aircraft near high pressure lines, airports or areas with severe magnetic interference.

Important Safety Information

Operation Handle the aircraft with care. It contains sensitive electronic components. Drops, damage, or water exposure may cause malfunction. Do not use a damaged aircraft (e.g., broken propellers).

Maintenance Do not disassemble or repair the aircraft yourself. This may cause damage or injury. If damaged or exposed to liquid, contact HUBSAN or an authorized dealer.For after-sales support: www.hubsan.com

Battery: Do not disassemble, squeeze, impact, burn, drop, or step on the battery.
Do not short-circuit the battery or touch terminals with metal.
Do not expose the battery to temperatures above 60°C.
Only use HUBSAN original chargers.

Flight: Always prioritize safety of yourself and others.

Do not fly in severe weather.
Do not attempt to catch the aircraft mid-air.
Not recommended for users under 14 or beginners.
After landing: lock motors → disconnect aircraft power → turn off remote control.

⚠ Note:
This product is designed exclusively for personal entertainment. Commercial, professional, or for-profit use is prohibited without official aviation approval.

Read the Disclaimer and Safety Operation Guidelines before use.

Symbol Legend

⊘ Prohibited Operation
⚠ Important Notes
Usage Tips
Glossary/Reference

HUBSAN Smart Battery

● Long-term storage: Charge to ⩾7.4V. If unused for 2 days, it automatically discharges to 7.4V to extend lifespan.
● Use only HUBSAN dedicated chargers.
● Do not charge in flammable/explosive environments.
● Recharge every 3 months if stored for over 3 months.

⊘1.Do not disassemble or reassemble the battery.
2.Do not short-circuit the battery
3.Do not use near heat sources
4.Do not immerse in water or expose to moisture.
5.Do not charge near fire or direct sunlight.
6.Do not puncture, hammer, or step on the battery.
7.Do not drop or impact the battery.
8.Do not use if damaged or deformed.
9.Do not solder directly to the battery.
10.Do not reverse-charge or over-discharge.
11.Do not reverse-connect the battery.
12.Do not connect to mains or car cigarette lighter.
13.Do not use in non-specified devices.
14.Do not mix with disposable batteries.
15.Do not exceed specified charging time.
16.Do not place in microwave or high-pressure container.
17.Do not use or store in direct sunlight.
18.Do not use in high-interference areas.
19.Do not use below 0°C or above 45°C.
20.Do not discharge above 5C; avoid over-discharge.
21. Do not remove battery while powered on for charging.
22.Stop use if abnormal smell or malfunction.
23.Keep away from children.
24.Use specified charger only.
25.When minors need to use batteries, parents should provide detailed guidance on their usage according to the manual!

Contents

User Notice
Important Safety Information
Battery Usage Precautions

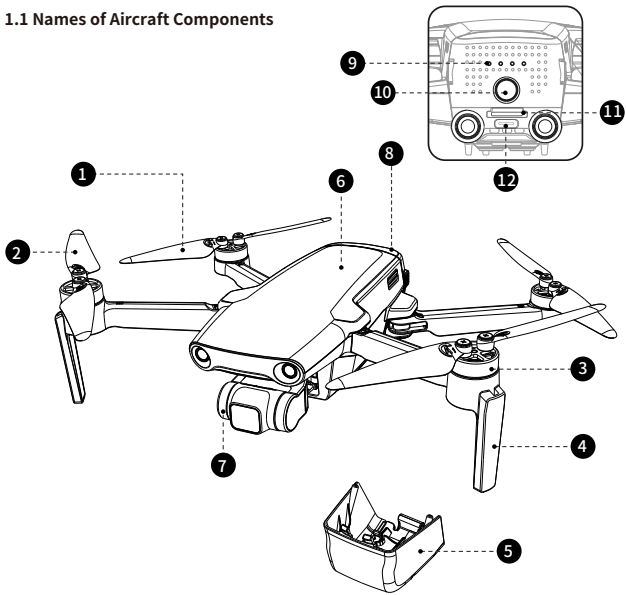
1 Aircraft----------1
1.1 Name of each component of the aircraft ---------- 1
1.2 Aircraft battery ---------- 3
1.2.1 Instructions for use of the smart charging tube house ---------- 3
1.2.2 Installation of the aircraft battery ---------- 4
1.2.3 Functions of the smart flight battery ---------- 4
1.2.4 Precautions for use in low temperature ---------- 5
1.2.5 Precautions for battery use ---------- 5
1.2.6 Storage of smart batteries ---------- 5
1.3 Propeller ---------- 6
1.4 Camera ---------- 6
2 Remote control ---------- 7
2.1 Name of each component of the remote control ---------- 8
2.2 Charging ---------- 8
2.3 Installation ---------- 8
2.4 Overview of the basic functions of the remote control ---------- 10
2.5 Angle of the remote control antenna ---------- 14
2.6 Calibration of the joystick and dial of the remote control ---------- 14
3 Operating system ---------- 15
3.1 Flight control interface ---------- 15
3.2 Steps to connect the aircraft ---------- 16
3.3 Compass calibration ---------- 17
3.4 Motor unlocking/locking ---------- 17

4 Aircraft Functions---------- 19
4.1 Flight Control Mode ---------- 19
4.2 Aircraft Indicators ---------- 20
4.3 Automatic Return to Home ---------- 21
4.4 Search for Drone Parking Area ---------- 22
4.5 Vision System and Infrared Sensing System --------- 22
4.6 Aircraft Heading Fill Light ---------- 24
4.7 Intelligent Mode ---------- 25
4.7.1 Creative Video ---------- 25
4.7.2 Orbital Flight ---------- 29
4.7.3 Waypoint Flight ---------- 30
4.7.4 Ray Flight ---------- 30
4.7.5 Follow Mode ---------- 31
4.7.6 Time-lapse Photography ---------- 32
4.7.7 Panoramic Photography ---------- 33
4.7.8 Movie Mode ---------- 34
4.9 Pan/Tilt Axis Adjustment ---------- 35

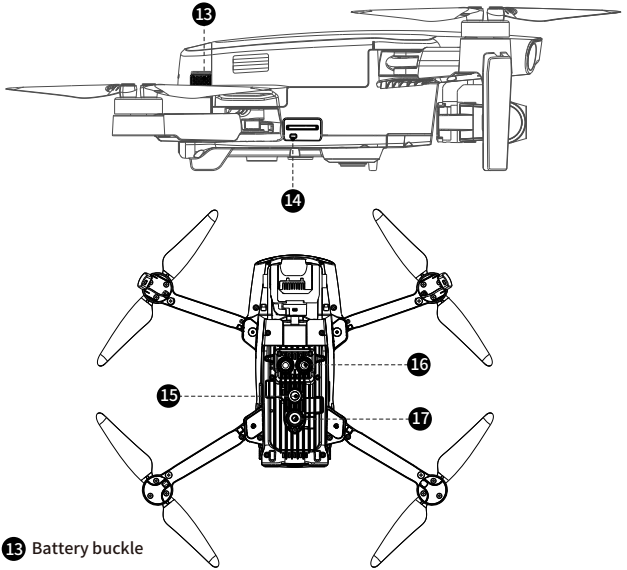
Frequently Asked Questions
Disclaimer
Suggestions

1.Aircraft

1.1 Names of Aircraft Components



1. Propeller A
2. Propeller B
3. Moto
4. Tripod (With Antenna)
5. Gimbal protection cover
6. Machine shell
7. Three - axis gimbal
8. Smart battery
9. Battery power indicator
10. light
11. Battery power switch
12. micro sd card slot



13 Battery buckle

14 Frequency pairing key

15 Downward vision

16 system

TOF assisted altitude

17 holding system

Notes: When the aircraft is in operation, it will generate heat.
After the flight, please avoid direct skin contact with the heat
sink at the bottom of the aircraft and the outer shell of the metal motor.

1.2 Aircraft Battery

1.2.1 Instructions for Smart Charging Butler

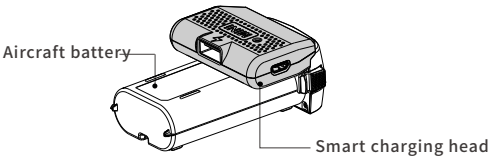
The aircraft uses a 2S smart lithium battery with a capacity of 3000mAh and a rated voltage of 7.2V. Consumers can charge it through a mobile phone charger that supports fast charging. It is recommended to use a 9V/2A charger, and it takes about 1.5 hours to fully charge.

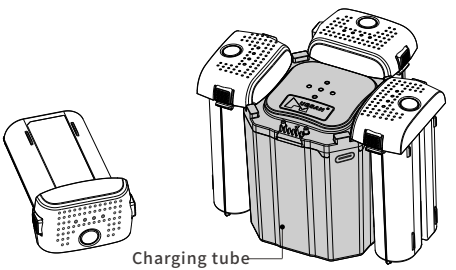
① Insert the TypeC charging cable into the charging butler. The light in the middle of the charging butler lights up, indicating that it starts working.

② With the battery powered off, insert the battery into the 4 interfaces of the charging butler. The charging butler blinks the indicator light for the interface that is charging, and the battery indicator light also blinks.

③ When the battery is fully charged, the battery power indicator lights go out, and the indicator light of the corresponding interface of the charging butler becomes always on, indicating that the battery is fully charged.

④ The charging sequence of the charging butler is to charge in order from the smallest number to the largest number on each interface.

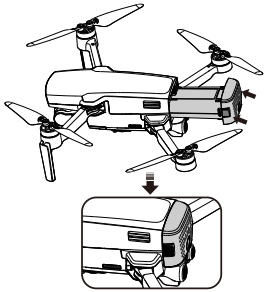




(Note: The charging manager is only available in the portable bag version not provided in the standard version.)

1.2.2 Installation of the Aircraft Battery

1. Prepare the aircraft battery
2. Put the battery into the battery compartment, press the left and right edge positions at the back of the battery (as shown in the figure) until the battery and the buckle position are completely embedded. Pay attention to listen for the sound of the buckle during the installation process
3. Check whether the top of the battery is level with the back of the aircraft to confirm that the battery is installed in place



1.2.3 Smart Flight Battery Functions

(1) Battery Activation: The smart battery needs to be connected to the APP for activation operation as prompted the first time it is used before it can be used. (2) Power Display: The battery is equipped with a power indicator light that can display the current battery power. (3) Upgrade Mode: The smart battery can perform firmware upgrades. When the battery is in the upgrade state, the two power indicator lights at the beginning and end are always on. (4) Automatic Discharge Protection During Battery Storage: After being fully charged and placed for 2 days without any operation, the battery will discharge to 7.4V (slight heating during this period is also a normal phenomenon) to protect the battery. (5) Automatic Balancing: Automatically balances the voltage difference between the internal battery cells to protect the battery. (6) Over-current Protection: High-current charging will seriously damage the battery. When the charging current is too large, the battery will stop charging. (7) Over-charging Protection: Over-charging will seriously damage the battery. The battery will automatically stop charging when it is fully charged. (8) Over-discharge Protection: Over-discharging will seriously damage the battery. When the battery is not in use, it will cut off the output when discharged to a certain voltage. The over-discharge protection will not be activated during flight. (9) Short-circuit Protection: It will cut off the output to protect the battery when a short-circuit is detected in the battery. (10) Cell Damage Detection: It will prompt that the battery is

Battery Switch	Short press to check the battery level, it will disappear after 3 seconds of display
	First press briefly and then press and hold for a long time to turn on/off the device. The indicator light gradually lights up/goes out

1.2.4 Precautions for Using at Low Temperature

(1) If the battery is used in a low-temperature environment (-10°C to 5°C), please fully charge it before use and warm the battery to above 10°C.
(2) Using the battery in an environment below -10°C will cause unforeseeable consequences.
(3) In a low-temperature environment, the actual discharge capacity of the battery is easily significantly less than the theoretical discharge capacity. When the APP prompts "low battery alarm", it is recommended to stop flying immediately.
(4) In a low-temperature environment, due to the limitation of the battery output power, the wind resistance of the aircraft will be reduced. Please operate with

1.2.5 Precautions for Battery Use

Before using the smart flight battery each time, be sure to fully charge it. The smart flight battery must be charged using the dedicated power adapter and smart charging head provided by HUBSAN official.
(1) After the flight, the temperature of the smart flight battery is relatively high. Wait until the battery cools down to room temperature before charging the battery.
(2) The allowable charging temperature range of the smart flight battery is 5°C to 40°C. If the temperature of the battery cell is not within this range, the battery management system will prohibit charging. The optimal charging temperature range is 25±3°C. Charging within this temperature range can extend the service life of the battery.
(3) When charging or installing the battery, do not hot-swap the battery. Hot-swapping the battery may cause irreversible damage to the battery. Therefore, when unplugging or plugging the battery from the charging head, charging manager,

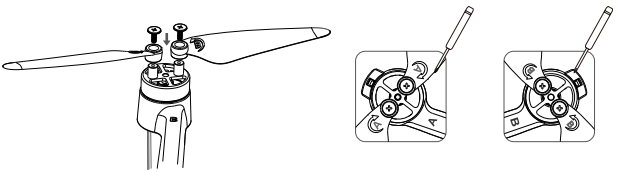
1.2.6 Preservation of Smart Batteries

Due to the physical characteristics of electronic components, even in the dormant state, smart batteries still have a weak current drain, which is regarded as a normal state. The drain current will continue to be lost until the battery's power is exhausted, and this behavior of draining power will cause irreversible damage to the battery's core. To avoid this situation, the following points need to be noted:
(1) Before storing the battery for a long time, please fully charge (or at least 70%) the battery before storing it.
(2) The battery needs to be taken out and recharged every six months to ensure that the battery power will not be drained due to the dormant drain current.
(3) The battery should be stored in a dry place, away from acids and alkalis, away from fire sources, without severe vibration, without pressure, and without flammable and explosive substances.
(4) The automatic discharge process of the battery will generate heat. Therefore, do not wrap the battery with materials with low thermal conductivity such as foam and fiber during storage, as this is likely to cause the battery to accumulate heat and burn out the core or even catch fire.

1.3 Propeller

Before installing the propeller, please carefully check the letters next to the motor of the aircraft and the letters on the propeller. The two sets of letters must be consistent during installation; take out the propeller, install it on the motor according to the letters, and then use a screwdriver to tighten the screw clockwise. When the blade is damaged or needs to be replaced, unscrew the screw counterclockwise to remove the propeller.

When replacing the blade, be sure to replace it with a new original blade screw to prevent propeller ejection. (The standard delivered accessories come with a new original blade screw)



1.4 Camera

The camera of MINI 7 adopts an ultra-high-definition image sensor.

1.When using the camera for the first time, please remove the protective film on the camera lens before use.

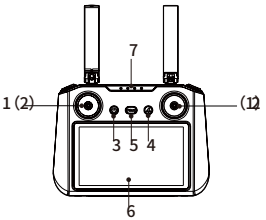
2.If you need to clean the lens during use, please use a professional optical dust-free cleaning cloth to gently wipe it.

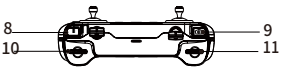
3.Please avoid soaking the camera in water.

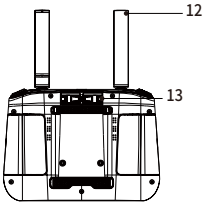
4.The normal working temperature range of the camera lens is 10°C to 55°C. When the temperature is lower than 10°C or higher than 55°C, it may cause the camera to be blurred or unable to output images normally

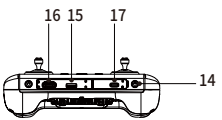
2. Remote control path

2.1 Names of various components of the remote control









Japanese hand mode:

(1) Throttle/left and right side flight control stick

(2) Forward/backward/steering control stick American hand mode:

1.Throttle/steering control stick

2.Forward/backward/left and right side flight control stick

3.Power switch

4.Long press one key to return to home (the aircraft returns to the take-off point)/short press to cancel return to home (when the aircraft is in the return to home state)

5.F/N/S mode: Movie mode/normal mode/sports mode

6.Display screen

7.Battery display

8.Multi-functional button

9.Take photo

10.Zoom wheel

11.Yuntai pitch control dial

12.Antenna

13.Joystick storage slot

14.Lanyard hole expansion interface

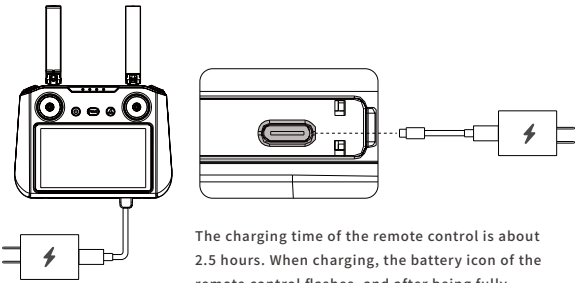
15.TF memory card slot

16.HDMI interface

17.Remote control charging/transferred wire interface

2.2 Charging

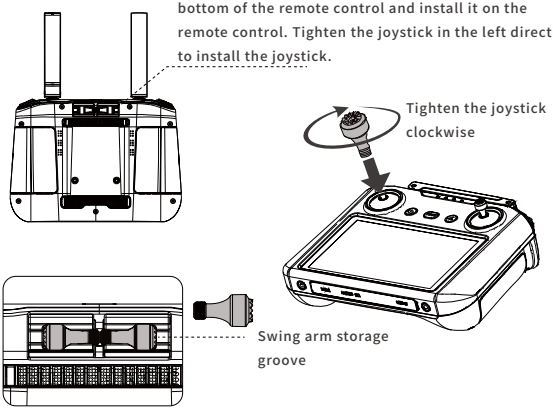
The remote control is charged using a Type C data cable, as shown in the figure:



The charging time of the remote control is about 2.5 hours. When charging, the battery icon of the remote control flashes, and after being fully charged, the icon is always on.

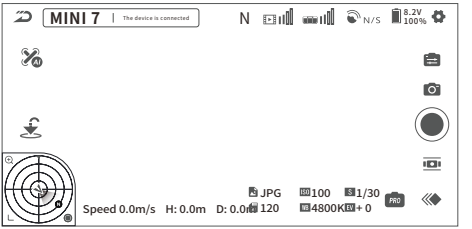
2.3 Installation

Take out the joystick from the joystick storage at the bottom of the remote control and install it on the remote control. Tighten the joystick in the left direction to install the joystick.

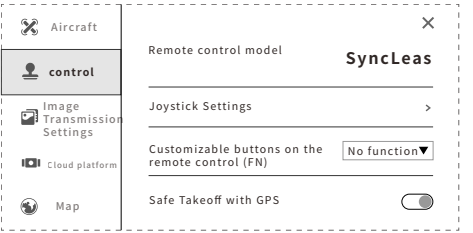


American hand mode and Japanese hand mode. The switching between the two operation modes can be set through the HUBSAN 3.0 APP.

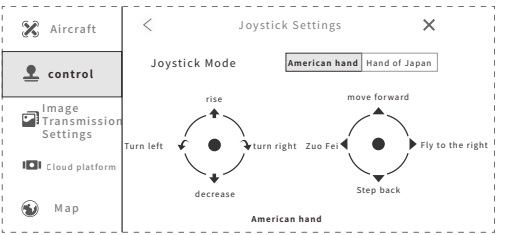
①Enter the flight control interface corresponding to the model of the aircraft in the APP,



②Enter the "Control" option from the settings interface and click "Stick Settings".



③Set the remote control joystick mode.

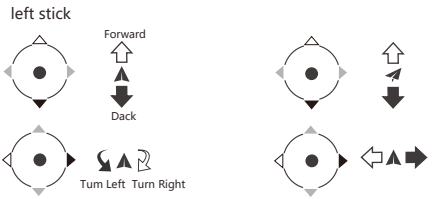


Basic operation of remote control

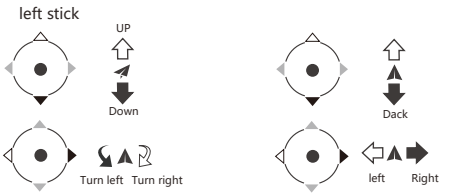
2.4 Remote Controller Features

	Key /Switch	Function
(1)	Throttle/ Rudder Stick	Move the stick forward or backward to ascend or descend the drone.Move the stick left or right to rotate the drone Clockwise or counter clockwise.
(2)	Elevator/ Aileron Stick	Move the stick forward or backward to fly the drone,forward or backward.Move the stick left or right to,fly the drone left and right.
1	Throttle /Aileron Stick	Move the stick forward or backward to ascend,or descend the drone.Move the stick left or,right to fly the Drone left and right.
2	Elevator/ Rudder Stick	Move the stick forward or backward to fly the,drone forward or backward.Move the stick,left or right to rotate the drone clockwise or counter clockwise.

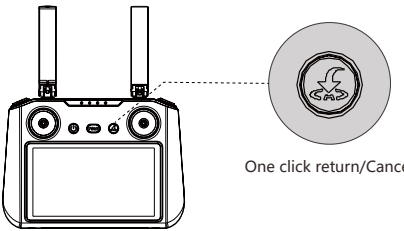
Japan Mode:



America Mode:

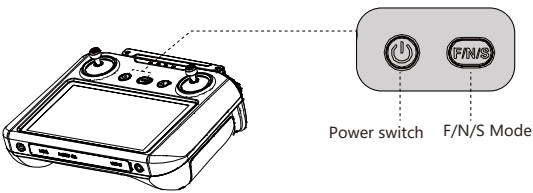


	Key /Switch	Function
3	One key return/-Cancel the returning home	Long press to enter the return mode (the aircraft,returns to the take-off point),short press the,button to exit One-key return (when the aircraft,is in returning status);During the returning,home,the remote control prompts "tick...Tick...,(one time/per 2 seconds),Note:the-One key return function on the remote,control can only use when the aircraft GPS is,normal and 6 or more satellits are found.

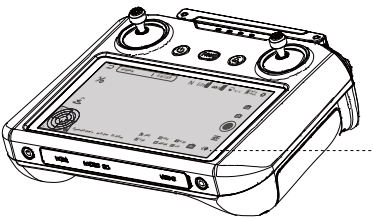


One click return/Cancel return

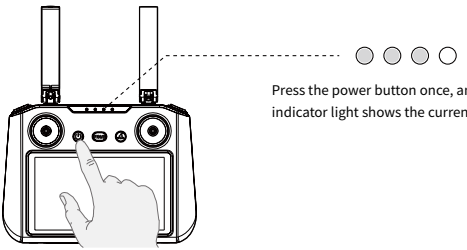
	Key /Switch	Function
4	power switch	Short Press,then press hold to power on or off
5	F/N/Smode (Movie/ Normal/Sport mode)	Movie mode: The aircraft flies at the set speed according to the APP. Normal mode: The aircraft flies at the maximum speed set by the APP. Sports mode: The aircraft flies at the maximum speed set by the APP



	Key /Switch	Function
6	Display screen	Display flight parameters

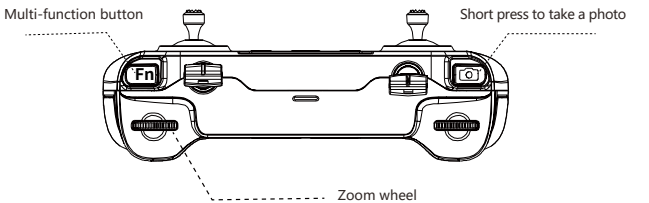


	Key /Switch	Function
7	Battery display	Four lights are power indicator lights, each representing 25% power
Low Battery Warning		Low battery warning for remote control:Remote,control "Beeps" <25%,di~di~,interval 10s, <15%,di~di~,interval 2s, <10%,didi~didi~,interval 1s

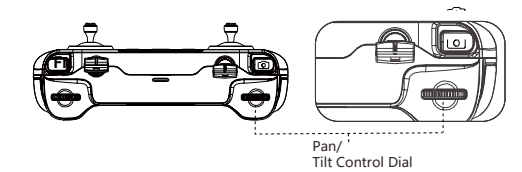


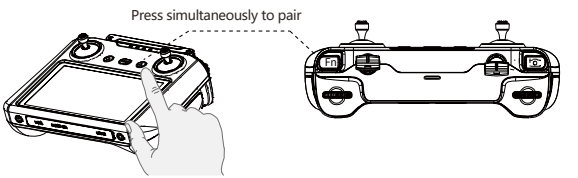
Press the power button once, and the power indicator light shows the current power.

	Key /Switch	Function
8	Multi-function button	Customizable multifunctional button
9	Take a photo	Short press to take a photo
10	Zoom wheel	Swipe the zoom dial to perform digital zoom

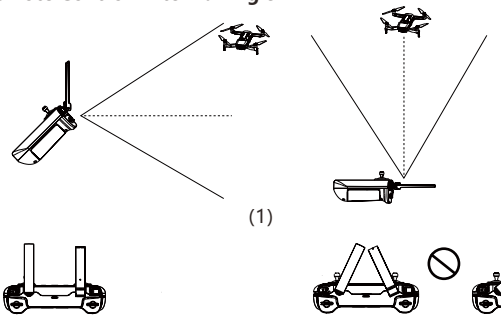


	Key /Switch	Function
11	Pan/ Tilt Control Dial	Control the pitch angle of the aircraft gimbal
Vehicle Code Matching		Press the return key and the multifunctional key simultaneously for 5 seconds to activate code pairing





2.5.Remote Control Antenna Angle

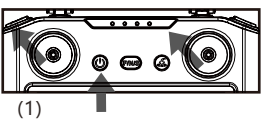


(1) Adjust the angle of the remote control antenna to align it as closely as possible with the direction of the aircraft's flight;
(2) When the aircraft is in a long-distance state, the distance between the two antennas should be appropriately tightened to maintain antenna directionality;
(3) Antennas should not obstruct or intersect with each other.

2.6.Control Sticks & Dial Calibration

Enter:

(1) Move both joysticks to the upper left corner simultaneously, then hold down the power button until the remote emits a "beep beep beep..." sound, indicating that it has entered calibration mode. Release the power button during the calibration process.

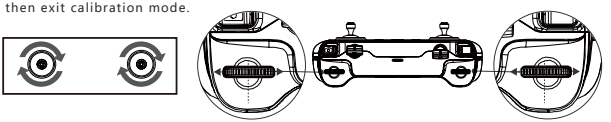


Calibration:

(2) Rotate both joysticks through their full range of motion three times each, then release them.
(3) Move the gimbal tilt wheel and zoom wheel to both ends of their range three times, then release all wheels.

Exit:

(4) Long-press the return-to-home button until the remote controller stops emitting the "beep beep beep..." sound, indicating the end of calibration. The remote controller will then exit calibration mode.



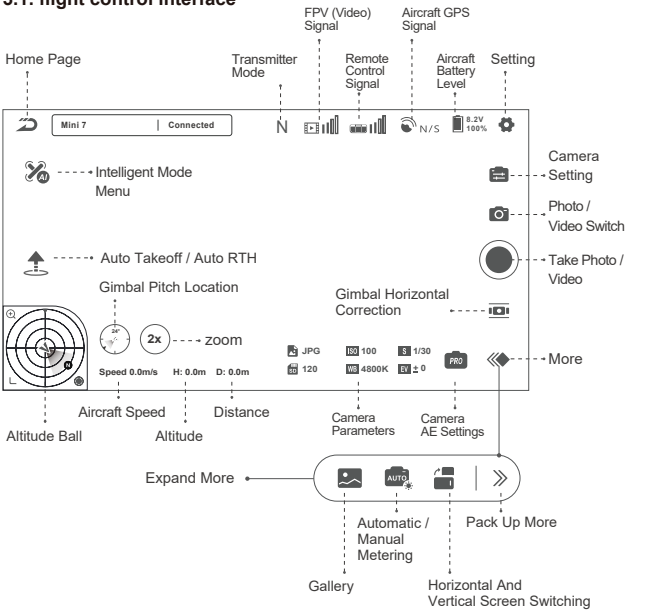
3. Operating System

Android 12.0

9. TF Card

(1) Support TF card, U3 above 16G/32G/64G/128G ;
(2) Hot plug is supported. You can insert the TF card even if drone is powered on. But it is recommended to plug in the TF card when drone is powered off.

3.1. flight control interface



(Note: You need to turn on the "longitude and latitude coordinates" switch in the settings and then can get the aircraft GPS coordinates and the transmitter GPS coordinates.)

3.2 Steps for connecting the aircraft

When using the aircraft for the first time, you need to activate the aircraft first

①Turn on the battery switch of the aircraft to power on the aircraft.

②Turn on the remote control switch and connect to the mobile phone, and wait for the remote control to successfully connect with the aircraft.

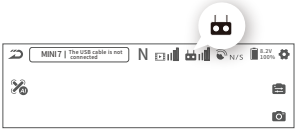
③Open the mobile phone HUBSAN 3.0 APP, and the activation interface will automatically pop up.

④New users can register an account as the activation account, and old users who have logged in to the APP before can directly enter the password to activate it.

(1) Open the HUBSAN 3.0 APP and select the model of the aircraft.

(2) Press briefly and then hold down to turn on the power of the aircraft.

(3) Press briefly and then hold down to turn on the power of the remote control, and



Before connection

After connection

(4) Enable frequency pairing mode (this step is only triggered manually by the user when pairing for the first time or replacing the remote control.)

① Press and hold the body frequency pairing key, and the aircraft enters the frequency pairing mode.

② Press and hold the FN + record button on the remote control to enter the frequency pairing mode, and bring the aircraft close to the remote control to complete the code pairing.

Tips:

(1) The frequency pairing between the aircraft and the remote control has been completed at the factory, and users can fly directly.

(2) After the aircraft enters the frequency pairing mode, the image transmission indicator light (green) flashes quickly. After the frequency pairing is successful, the image transmission indicator light (green) will become always on.

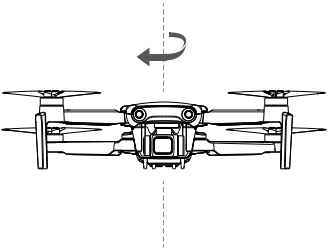
(3) During frequency pairing, please keep the distance between the aircraft and the remote control within 1 meter.

(4) The frequency pairing cannot be activated during the flight of the aircraft.

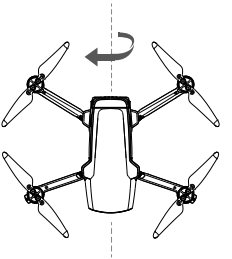
3.3 Compass Calibration

When used for the first time, the compass calibration will pop up automatically before takeoff. According to the interface prompt, first rotate the aircraft in the horizontal direction, and then rotate the aircraft in the vertical direction. After completion, the calibration pop-up window disappears. The first flight must pass the magnetic calibration before taking off.

The compass is easily interfered by other electronic devices, resulting in abnormal flight data. Regular calibration can keep the compass in the best state. After entering the APP settings interface, select magnetic calibration.



Compass 1



Compass 2

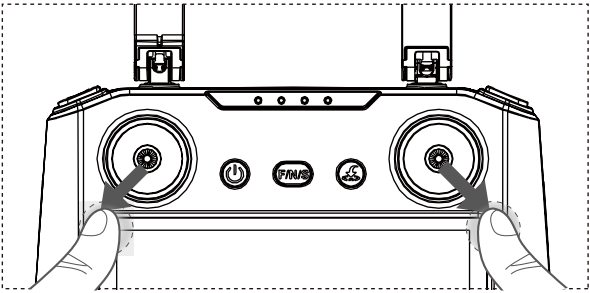
3.4 Motor Unlock/Lock

Motor Startup Conditions

(1) The aircraft, remote controller, and mobile phone have been successfully connected; (2) The aircraft has been calibrated by the compass (the APP does not prompt that the compass needs to be calibrated);

(3) The aircraft must be placed horizontally;

(4) When the GPS signal has not reached the unlocking condition, it is recommended not to force takeoff;



Start the motor

Start the motor with the external eight-pole lever. After the motor starts, please release the rocking rod

Stop the motor

When the motor is running, the outer eight striking rod stops the motor. After the motor stops, please release the operating lever.

Forcibly stop the motor

When flying in the air, pulling the outer eight stick for 2 seconds can forcibly stop the motor. This function should be used with caution as it may cause the drone to fall and potentially endanger personal safety. This function is only allowed to be used by default in case of malfunctions (such as abnormal situations like rolling). If it needs to take effect at any time, the user needs to manually set it in the HUBSAN 3.0 APP.

4 Vehicle Function

4.1 Remote Control Mode

Fly control mode (Fly control judges GPS signals and switches to corresponding modes)	GPS mode	Use GPS and down-looking vision system to realize precise hover, stable flight, intelligent flight mode, etc. The vision system needs to work in a well-lit environment.
	Light Flow Mode	When the aircraft hovers at low altitude or flies at low speed slowly, it is in the optical flow mode. This mode is only effective indoors and when the height is less than 5 meters. When it is greater than 5 meters and flying outdoors, GPS mode or attitude mode is adopted. In the optical flow mode, the maximum flight speed of the aircraft is 2m/s.
	Posture mode	When the GPS signal is poor or the compass is interfered and the visual positioning working conditions are not met, the aircraft will enter the attitude mode, which only supports manual flight and prohibits the use of the intelligent mode. In the attitude mode, the aircraft has abnormal positioning. Please land as soon as possible.

Flying speed	Movie Mode	Maximum speed 3m/s
	Ordinary mode	Maximum speed 8 m/s, and the maximum speed can be adjusted from 10% to 100% in the APP settings
	Sports mode	The maximum speed is 16 m/s, and the sports mode can only be entered in GPS mode
	Endurance time	30 minutes

4.2 Aircraft Indicator

Video Indicator (Red)、Image Indicator (Green) Functions Indicator	
Camera Error	Red LED flashes slowly (1 time/second)
Function Error	Red LED、Green LED off
Booting	Red LED、Green LED off
Video Indicator Working	Red LED solid
Image Indicator Working	Green LED solid
Binding mode	Green LED flashes quickly
Disconnected	Green LED flash quick and slow
Upgrading	Red&Green LED from flashes quick to slow (Flashing quickly when transferring files, and slow flashing when writing flash)

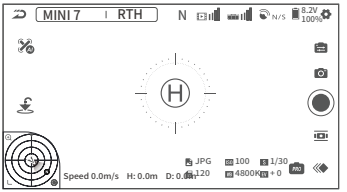
4.3 Return-to-home(RTH)

There are three types of RTH: One-key RTH, Low Battery RTH, and Failsafe RTH. When taking off, and GPS has locked more than 6 satellites, the aircraft will mark take off point as home point. If there is no GPS signal and drone is forced to take-off, Whenever GPS locks more than 6 satellites, it will record that point as home position.

RTH-Process
1. Ensure the "home" point is recorded. 2. Press the return to home icon 3. The aircraft will adjust its direction. 4. The aircraft will climb to the safe altitude as set in the App (1) Regardless of the current height of the aircraft, if the aircraft is within 5 meters of the horizontal distance from home point, it will land; (2) When the horizontal distance is greater than 5 meters, the aircraft will rise to the set altitude and then return. 5. Search for the Aircraft parking apron during the landing process (This feature must be turned on in app before taking off and or landing)
One-key RTH
APP one-Key RTH / remote control one-Key RTH
Low Battery RTH
1. The aircraft will automatically Return to home according to its own power calculations and the distance from the Home Point. 2. Aircraft will land automatically at the same point upon low battery RTH activation, If there is no GPS signal or weak GPS signals 3. When taking off, if it is forced take off with weak GPS signals or GPS signals are not good, the aircraft will automatically return to the place where it achieved, good GPS signal for the first time. 4. When the aircraft battery power percentage is lower than 10%, the aircraft will start landing automatically if there is no joysticks operation. We can cancel it by using the operate the transmitter joystick, and let the aircraft land slowly or climb again. 5. If the battery level is lower than 1%, the aircraft will forced to landing, which cannot be manually canceled or forced to climb again.
Failsafe RTH
When the aircraft loses connection with the remote control for more than 5 seconds, the aircraft automatically returns or land directly. Performance requirements: (1) After the aircraft loses control for 5 seconds, it triggers automatic return; (2) If the drone reconnects, it will continue to perform the returning procedure when Failsafe RTH mode is activated; (3) Landing directly when there is no GPS signal or the signal is not strong.

4.4 Landing Apron Search

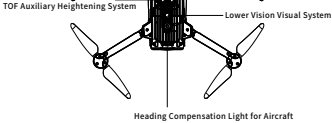
When the aircraft is landing or returned to a height of about 10 meters from the ground, it will automatically enter the search for the drone apron function.



(1) The gimbal camera points downward to optically search for the Aircraft parking apron [H], Apron requirements: 1 Sharp contrast, 2 white "H" lettering,
3 Apron without obstruction.
(2) Once drone locks the apron visually it will descend smoothly on apron. When landing to a height of less than 3 meters, the aircraft camera will switch to forward view and no longer adjust the position of the aircraft. From this altitude aircraft will land quick. If an error is identified or other unexpected circumstances occur, press the stop button to exit the function.
(3) If aircraft can' t find the parking apron or battery is critically low, the aircraft will land directly.
(4) The search for the Aircraft parking apron function cannot be performed if video is being recorded.
(5) If you do not need to use this feature, please turn it off in app.

4.5 Vision Systems and Infrared Sensing Systems

ZINO MINI 7 is equipped with Forward, Backward, and Downward InfraredSensing Systems, providing omnidirectional obstacle sensing.The main componentsof the Forward, Backward,and Downward Vision Systems are two cameras locatedon the nose, rear side, and the underside of the aircraft. Lateral Vision Systems aretwo cameras located on either side of the aircraft.The main components of the Upward and Downward Infrared Sensing Systemsare two infrared modules located on the upper and underside of the aircraft,The Downward Vision System and Infrared Sensing System helps the aircraft maintainits current position and hover in place more precisely.The Auxiliary Bottom Light located on the underside of the ZINO MINI 7 aircraftimproves visibility for the Downward Vision System in weak light conditions.



Observation range

Downward vision

Precise hovering range: 0.5m-5m, visual hovering range: 0.5m-5m

The downward perception system can capture and analyze data about the surrounding environment by fusing multiple sensor data. By combining TOF technology, ground obstacles can be identified in real time, enhancing its self-protection ability in complex environments.

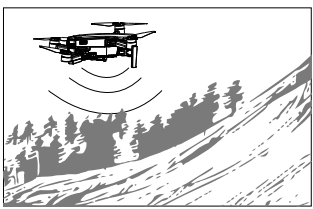
Real-time obstacle detection: TOF sensors can monitor the ground and surrounding environment below the drone in real time, detect the presence of obstacles, and measure the distance.

Dynamic avoidance decision: Once an obstacle is detected, the system will quickly calculate the appropriate avoidance path. The drone can avoid obstacles by adjusting the flight altitude.

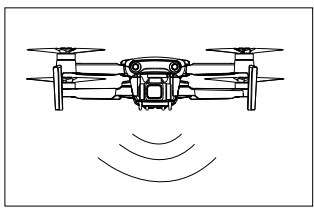
Accurate positioning and navigation: TOF sensors combined with visual perception systems can provide more accurate positioning information, helping drones maintain stable flight in complex terrain and avoid collisions with obstacles.

Usage scenarios

The positioning function of the downward vision system is suitable for environments with no GNSS signal or poor GNSS signal, and the optimal working height range is 0.5-5m. The downward vision perception system is automatically turned on in normal or movie mode.



Perceive obstacle distance



Precision hovering

Note: (1) The optical flow vision system can only assist flight when the surrounding light is sufficient and the texture is rich. It cannot completely replace the user's judgment. Please pay attention to the aircraft situation and APP prompts, and do not over-rely on the optical flow vision system.

(2) The optical flow vision system cannot be used normally in scenes such as too bright or too dark ambient light, mirrors, water surfaces, reflective surfaces, and sparsely textured surfaces.

(3) The optimal working range of the downward optical flow vision system is 0.5-5 meters. When it exceeds this range, the positioning performance of the downward optical flow vision system may fail. Please fly with caution.

(4) Please ensure that the optical flow vision system lens is clear and do not block or interfere with the optical flow vision system.

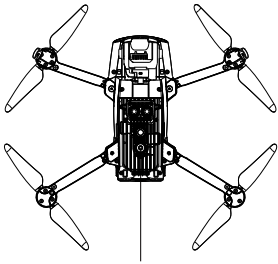
(5) In the absence of GPS or weak GPS environment, the bottom vision system can provide auxiliary positioning; when the outdoor GPS is good, the drone will automatically switch to GPS positioning mode.

⚠ Please pay attention to the flight environment. The visual system and infrared sensor system can only play a safety auxiliary role under limited conditions and cannot replace human judgment and control. Users should always pay attention to the surrounding environment and APP-related warnings during the flight, maintain control of the aircraft throughout the flight and be responsible for the control behavior.
Without GNSS, when using the visual system in an open and flat field, the maximum hovering height of the aircraft is 5m. The optimal working altitude range of the visual positioning system is 0.5-5m. When flying beyond this range, the visual positioning performance may decrease. Please fly with caution.
· In the APP, you can set the opening mode of the fill light (on, off or automatic). If you choose automatic, the fill light will automatically turn on when the ambient light is insufficient. At this time, the visual positioning performance will decrease. If the GNSS signal is poor, please fly with caution.

⚠ The visual system may not work properly on the water surface. Therefore, when the landing function is triggered, the aircraft may not actively avoid the water below. It is recommended that users maintain full control of the flight and make reasonable judgments based on the surrounding environment, and do not rely too much on the visual system.
· The visual system cannot recognize surfaces without texture features and cannot work properly in environments with insufficient or excessive light intensity. The visual system cannot work properly in the following scenarios:
1. Pure color surface (such as pure black, pure white, pure blue, pure red, etc.);
2. Surface with severe reflection (such as water surface, ice surface, glass curtain wall, smooth surface, etc.);
3. Transparent surface (such as transparent glass, etc.);
4. Scenes with drastic changes in lighting (such as backlight, etc.);
5. Extremely dark or extremely bright scenes;
6. Surfaces of materials that have strong absorption or more reflective effects on infrared (such as mirrors, etc.);
7. Surfaces of moving objects (such as above the flow of people, above the grass blown by the wind, etc.);
8. Surfaces with sparse textures;
9. Surfaces with highly repeated textures (such as repeated squares of the same color, etc.);
10. Small obstacles;
11. The safe operating temperature of the visual system is 0-50 degrees. If the temperature exceeds this range, the stability of the visual system will decrease. Please try to avoid using the visual assistance system outside the safe temperature.
· Do not interfere with the visual system in any way and ensure that the lens is clear and spotless.
· If the aircraft is hit hard, the visual system needs to be recalibrated.
· Avoid flying in rainy, foggy weather or other low visibility (visibility less than 100m).
· Please check the surface glass of the infrared sensor system and visual system before takeoff: a.Remove the film, stickers, and other obstructions on the surface. b.If there are water droplets, fingerprints, dirt, etc., please wipe them clean first (please use a dust-free cloth to wipe, do not use organic solvents such as alcohol). c.If the surface glass is dropped, broken, scratched, worn, etc., please return it to the factory for repair.
· Do not block the infrared sensor in any way.

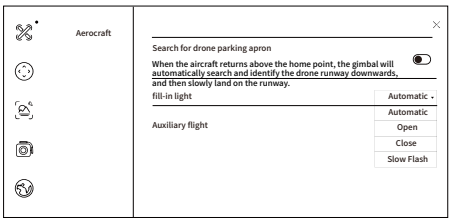
4.6 Aircraft heading fill light

The bottom of the aircraft is equipped with heading indicator lights. When flying at night or in low visibility weather conditions, the lighting system should be turned on and ensured to be in good working condition.



Attention: When flying at night, please follow local air traffic control regulations

The heading fill light is turned on as shown in the following figure: enter the APP settings, aircraft bar, slide to "fill light" on, turn on the fill light, which is divided into three light modes: - automatic (automatic mode) - on (constant light mode) - slow flashing (flashing mode). When flying at night or in low visibility, turn on the heading fill light. Within the line of sight range, you can quickly confirm the position of the aircraft.



4.7 Smart mode

4.7.1 Creative Video

360° Shooting :

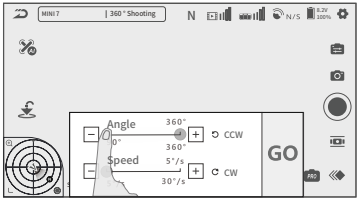
(1) Select the direction of rotation (clockwise / counterclockwise);

(2) Set the rotation angle, the range of rotation angle (90 ° -360 ° ,accuracy 1°);

(3) Set the rotation speed (1-30 ° / sec, accuracy 1);

(4) Click GO, the aircraft rotates in place hovering at its position, shooting a video during the rotation;

(5) you can click the exit button any time to exit the mode and save the video.



Comet Mode:

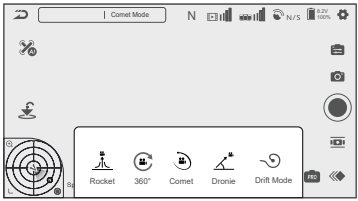
After selecting the center point on the APP, the aircraft will automatically fly along the eclipse track and record video. Please only use it in an open area.

APP operation:

(1) Select the target, the aircraft camera is always aimed at the selected target when shooting

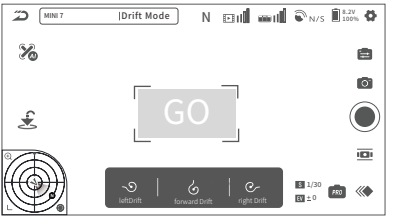
(2) Select the flying direction (clockwise/counterclockwise)

(3) Click GO, the aircraft starts recording the video and performs eclipse flight move.



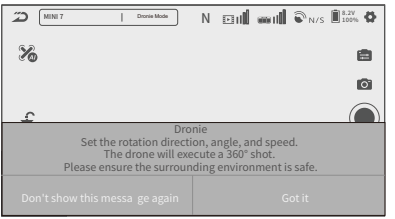
Drift Mode:

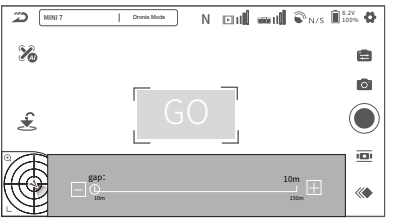
The drone takes advantage of its high maneuverability to perform a smooth, car-like "drift" rotation by automatically and precisely adjusting its direction and movement. While creating a dynamic spinning effect, the drone keeps flying smoothly along the path. It also records timelapse footage during the flight and automatically edits it into a cinematic video — all with just one tap.



Dronie Mode:

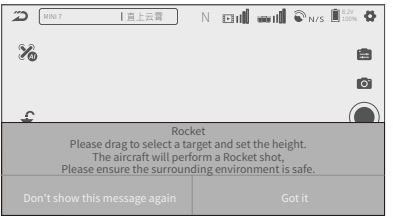
After intelligently locking onto the selected subject (person, scenery, etc.), the drone gradually flies backward while automatically recording video. This distance change showcases the spatial layers of the scene and the relationship between the subject and the environment. It is mainly used for aerial photography, short video creation, and landscape documentation.

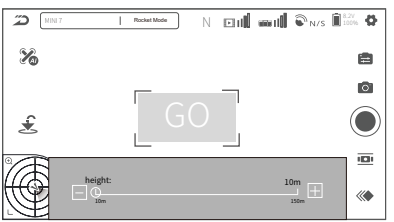




Rocket Mode:

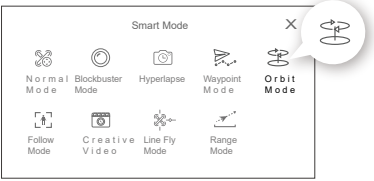
Based on the selected subject and its distance, the drone intelligently locks onto the target and ascends automatically along a set trajectory. While climbing, it captures video from a "God' s-eye view," keeping the subject centered in the frame. This mode is ideal for aerial scene capturing and landscape showcasing.





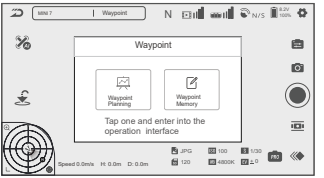
4.7.2 Orbiting

Tap on "Mode Selection" then "Orbit Mode" to set the current location or the position of the mobile device as the center. During Orbit mode, you can adjust the speed and direction by moving the control stick left and right and adjust the orbit radius by moving the control stick forward and backward.



4.7.3 Waypoint

The aircraft will fly along the set flight path drawn on-screen or saved ,you can adjust the speed of the aircraft during flight .



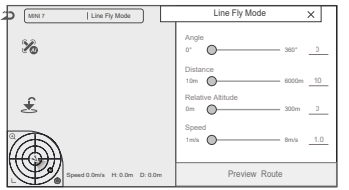
Waypoint Planning: You can preset the waypoint parameters such as the number of waypoints, the altitude of each waypoints and other parameters. The aircraft will follow the preset parameters after you activate waypoint mode. You can control the flying speed during flight or you can pause or resume the Waypoint mode.

Waypoint Memory: After entering the mode, fly the aircraft and tap on "Memorize Waypoints" on the APP and the aircraft will mark the location. After memorizing all the waypoints, upload and execute, aircraft will fly according to recorded waypoints.
If the aircraft is not in the starting position, the aircraft will fly to the starting waypoint position before starting this function.

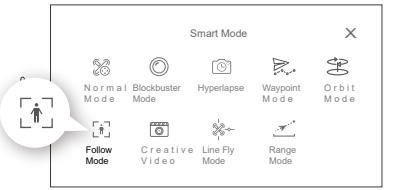
4.7.4 Line Fly Mode

Tap on "Mode Selection" then "Line Fly Mode".Set the angle,distance and speed of the aircraft.

Performance requirements:
1. Set the angle (0 ~ 360 °, accuracy 1 °);
   Set the distance (10-6000 meters, accuracy 1 meter);
   Set the speed (1-8 m / s, accuracy 0.1);
2. Once setup is complete, the aircraft will execute Line Fly mode.
During the flight, pilots can take photos or videos manually , or pause / resume / stop the flight at any time.
During the flight, you can control the altitude only.
3. You can pause/resume/stop the flight at any time.
4. When the aircraft is low on battery or lost connection from radio, aircraft will always return to home.



4.7.5 Follow Mode



Active Tracking 3.0

MINI 7 Active Tracking 3.0 includes Spotlight, Active Tracking, Circle Tracking.

Spotlight: The aircraft remains in the current position, and only adjusts the heading and gimbal camera to lock the target in the middle of the screen. Use the stick to control the movement of the aircraft: the roll stick controls the aircraft to rotate around the target, the pitch stick controls the aircraft to approach or move away from the target, the throttle stick controls the aircraft height, and the yaw stick controls the composition.

Active Tracking 3.0 includes the following modes: Trace and Parallel.

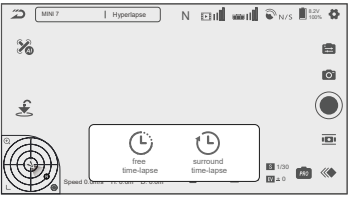
(1)The aircraft tracks the subject at a constant distance and follows the target as it moves. The maximum flight speed is 8m/s.

(2)The aircraft tracks the subject at a constant angle and distance from the front and side. The maximum flight speed is 8m/s.The parallel modehas no available obstacle avoidance.

Circle Tracking: The aircraft flies around the target with a set radius and speed and supports static and dynamic targets (dynamic targets refer to people, vehicles, and ships, and the speed should not be too large, and the following height will not change with the dynamic target). The maximum circle speed is 5m/s.

4.7.6 Hyperlapse

When you choose Hyperlapse mode, you can choose free and Circle.

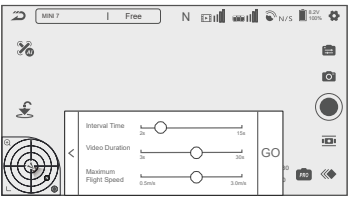


The aircraft will take a certain number of photos and compose time-lapse videos automatically according to the parameters set. During the process of shooting, the user can control the flight of the aircraft freely.

(1) choose free;

(2) set the shutter interval, the video length, and the maximum flight speed;

(3) after done, please click ‘GO’ and start shooting.



Circle

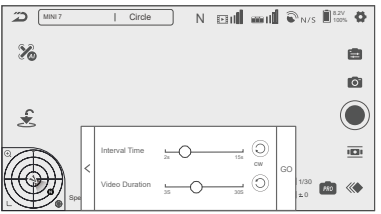
The aircraft will take a certain number of photos and compose time-lapse video automatically according to the circle point of interest and the parameters set. During shooting, moving any sticks on controller will automatically quit the task.

(1) select circle.

(2) set the shutter interval, video length and maximum flight speed.

(3) set the circle direction, and position the circle center by adjusting the circle radius and the direction of the aircraft nose.

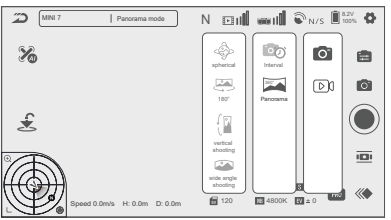
(4) click ‘ GO’ and start shooting.



4.7.7 Panorama mode

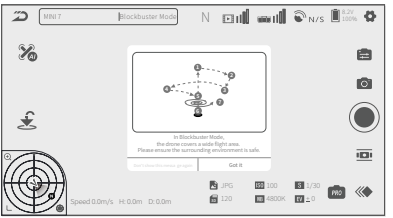
Set the photo mode in the camera settings to panoramic photo,you can choose spherical, 180°, vertical shooting and wide angle shooting. In this mode, the aircraft will perform the shooting task automatically. After the shooting done, you can view photos and videos from the SD card, also export photos and videos and compose them. You can quit this mode any time during shooting.

Note: Aircraft will not stitch and compose the final panoramas or spheres, you will need to use after market software to perform the stitching.



4.7.8 Blockbuster Mode

The drone intelligently locks onto the target based on its type and distance, automatically performing classic aerial shots with close-up and long-range flight paths. After shooting, the footage is auto-edited into a video. Even beginners can create a blockbuster with one click.

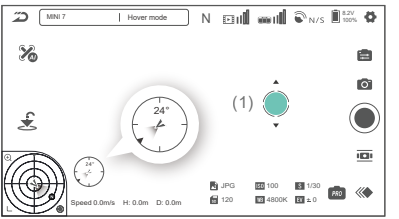


4.8 Gimbal Pitch Adjustment

- Please make sure there are no stickers or impurities on the gimbal before take off, and place the aircraft on flat and open ground. Please do not bump the gimbal after the power is turned on.
- The gimbal contains precision parts. If it is hit or damaged, the precision parts will be damaged, which may cause the performance of the gimbal to decrease. Please take good care of the gimbal and camera from physical damage.
- Please keep the gimbal clean and avoid the gimbal from contacting foreign objects such as sand or stone, otherwise it may block the movement of the gimbal and affect its performance.
- If the aircraft is placed on uneven ground or grass, the ground object touches the gimbal, or the gimbal is subjected to excessive external force (such as being bumped or broken) may cause the gimbal motor to be abnormal.
- Do not add any objects to the camera head, otherwise it will affect the performance of the head and even burn the motor.
- Remove the gimbal protection latch before use and then turn it on. Reinstall the gimbal latch to protect the gimbal during storage or transportation.
- Flying in heavy fog or clouds can cause the gimbal to condense, resulting in temporary failure. If this happens, the gimbal can return to normal after drying.

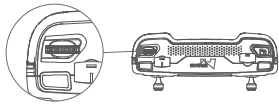
APP Adjustment

long press the blank of the video preview interface and the mobile device will vibrate once. When appears (1) at the position which you press, you can adjust gimbal pitch by sweeping your finger up and down the screen.



Transmitter Adjustment

You can adjust the angle of gimbal pitch by fiddling with the control thumb-wheel of gimbal pitch.



FAQ

1.The mobile device and remote control cannot be connected

(1) Check whether the status of the APP control signal icon has changed

(2) For USB Settings on android phones, check out "Android phone connection tutorial"

2. FPV transmission freezing, gets stuck or easy to disconnect

(1) Adjust the antenna Angle and point vertical side of the antennas to the aircraft, without anything blocking between antennas

(2) Change the flight site, please do not fly near tall buildings, or near signal tower

(3) Update the latest firmware

3.Abnormal aircraft hovering

(1) Change the flight site, please do not fly near tall buildings and signal towers

(2) Do the compass calibration and horizontal calibration

(3) Wind is too strong to affect the flight

(4) If the aircraft are indoors, it is recommended to move to the outside for flight

4. Aircraft GPS accuracy is not precise or cannot pass the GPS accuracy test

(1) in the open area of the outdoor with GPS over 6 satellite

(2) Walk around near the aircraft

(3) Replace the mobile device

5.The battery cannot charge

Re-plug the charger and battery

6. short flight times

Battery overcharge or high temperature environment can easily lead to reduction of battery life, it is recommended to keep the remaining 60% or so, full charge again before use

7. The tilt Angle of the gimbal is too large or the gimbal show abnormal behavior

(1) Restart the aircraft and re-calibrate the gimbal

(2) Check on the APP to see whether the gimbal status is normal

8. Gimbal initialization fails

Before starting the aircraft, take off the protective cover of the gimbal and make sure nothing blocks gimbal movement.

9.The picture is not clear

(1) Check whether the lens protective film is peeled off

(2) Please shooting in a environment with good lighting

(3) Shooting parameters setting in camera Settings of APP

10. Lens fogging

(1) humid climate causes lens fogginess, change aircraft storage location

(2)Place some desiccant in the protective cover of the gimbal when storing

11. Picture or video is lost

Perform completing recording operation after recording video, otherwise it may lead to video damage or loss

Disclaimer

HUBSAN accepts no liability for damages, injuries or any legal responsibilities directly or indirectly from the use of HUBSAN products under the following conditions:

1.Damages, injuries or any legal responsibilities when users are drunk, under the influence of drugs or anesthesia, dizzy, fatigued, nauseous and / or affected by other conditions both physical and mental that could impair sound judgment and / or personal ability.

2. Subjective misjudgment and / or intentional misoperation of products.

3. Any and all mental damage, trauma, impairment, illness, compensation caused / solicited by accidents involving HUBSAN products.

4. Product operation in no-fly zones (i.e. natural reserves).

5. Malfunctions or problems caused by modification, refit, replacement or use with non-HUBSAN accessories / parts, failure to follow the guidance of the manual in assembly or operation.

6.Damages, injuries or any legal responsibilities caused by mechanical failures due to natural wear and tear (aircraft flight time clocking in 100 hours or above), corrosion, aging hardware, etc.

7. Continued flight after low voltage protection alarms are triggered.

8. Deliberately flying aircraft under abnormal conditions (such as when water, oil, soil, sand or other unknown material are inside the aircraft and / or transmitter are incompletely assembled, the main components have obvious faults, obvious defect or missing accessories, etc).

9. Flying in the following situations and / or environments: areas with magnetic interference (such as high voltage lines, power stations, broadcasting towers and mobile base stations), radio interference, government regulated no-fly zones, if the pilot loses sight of the drone and suffers from poor eyesight or is otherwise unsuited for operating HUBSAN products.

10. Aircraft use in or exposure to bad weather, such as a rain, wind, snow, hail, lighting, tornadoes and hurricanes.

11. Products are involved in / exposed to collisions, fire, explosions, floods, tsunamis, man made and / or natural structure collapses, ice, avalanches, debris, landslides, earthquakes, etc.

12. The acquisition, through use of HUBSAN products (specifically but not limited to aircraft), of any data, audio, video that results in infringement of law and / or rights.
13. Misuse and / or alteration of batteries, product / aircraft circuits, hardware protections (including protection circuits), RC model and battery chargers.
14. Any malfunction of equipment or accessory, including memory cards, that results in the failure of an image or video to be recorded or to be recorded in a way that is machine readable.
15. Users who engage in reckless, unsafe flying (with or without sufficient training).
16. Non compliance with precautions, instructions, information and operation guidelines / methods given through official Hubsan website announcements, product quick start guides, user manuals, etc.
17. Other losses, damages, or injuries that are not within the boundaries of Hubsan responsibility.
RISK OF EXPLOSION IF BATTERY IS REPLACED BY AN INCORRECT TYPE.
DISPOSE OF USED BATTERIES ACCORDING TO THE LOCAL REGULATIONS.
HAZARDOUS MOVING PARTS KEEP FINGERS AND OTHER BODY PARTS AWAY.
Declaration of Conformity
Hereby, SHENZHEN HUBSAN TECHNOLOGY CO., LTD.,declares this product is in compliance with the essential requirements and other relevant
provisions of Directive 2014 / 53 / EU. A copy of the original Declaration of Conformity can be obtained at the following address: Unit 2801, Building F, Xinghe WORLD , Yabao Road, Bantian Street, Longgang District, Shenzhen , China.
This product bears the selective sorting symbol for waste electrical and electronic equipment (WEEE). This means that this product must be handled pursuant to European Directive 2012 / 19 / EU in order to be recycled or dismantled to minimize its impact on the environment.
For further information, please contact your local or regional authorities. Electronic products not included in the selective sorting process are potentially dangerous for the environment and human health due to the presence of hazardous substances.

FCC INFORMATION

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

• Reorient or relocate the receiving antenna.
• Increase the separation between the equipment and receiver.
• Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
• Consult the local dealer or an experienced radio / TV technician for help.Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment. This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with a minimum distance of 20cm between the radiator & your body. This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.

Electrical and electronic equipment that are supplied with batteries (including internal batteries)

WEEE Directive & Product Disposal

At the end of its serviceable life, this product should not be treated as household or general waste. It should handed over to the applicable collection point for the recycling of electrical and electronic equipment, or returned to the supplier for disposal. Internal / Supplied Batteries.

This symbol on the battery indicates that the battery is to be collected separately.This battery is designed for separate collection at an appropriate collection point.

Noise Test of MINI 7

Noise	Observation point	Suspend	Flight 1 m/s
	Ground observation point (directly below)	74 dB(A)	70 dB(A)
	Side observation point (equipotential plane)	64 dB(A)	61 dB(A)
	Note: Outdoor cement ground, ambient noise 44 dB		

Please read the operating instructions carefully before use!

- ●Never leave units unattended when charging
- ●Unplug the charging cable immediately after charging
- ●Propellers may cause injury
- ●This product is not a toy
- ●Not suitable for children under 14 years of age

Product name: MINI 7
Product Standard Number:Q / HBS 001-2017
Vendor: Shenzhen Hubsan Technology Co., Ltd
Address: Unit 2101-02C, Building F, Xinghe WORLD , Yabao Road, Bantian Street, Longgang District, Shenzhen
Email: service@hubsan.com
WWW.HUBSAN.COM